

ETHICAL HACKING

CO2-ASSESSMENT TOOL 2

Experiment 1: OSINT-Based Email Enumeration Using theHarvester (20 Marks)

Aim:

To perform an OSINT-based email enumeration assessment on the target domain **tesla.com** using theHarvester in Kali Linux.

Tool Used

- Kali Linux
- TheHarvester

COMANDS USED:

pwd

theHarvester --version

theHarvester 4.10.1

theHarvester -h

theHarvester -l

theHarvester -d tesla.com -b crtsh

theHarvester -d tesla.com -b crtsh -f tesla_report

ls

OUTPUT:

```
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)~$
$ pwd
/home/kali
(kali@kali)~$
$ theHarvester --version
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
* theHarvester
*
* theHarvester 4.10.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*
*****
usage: theHarvester [-h] -d DOMAIN [-l LIMIT] [-s START] [-p] [-s] [--screenshot SCREENSHOT] [-e DNS_SERVER] [-t] [-r [DNS_RESOLVE]] [-n] [-c]
                  [-f FILENAME] [-w WORDLIST] [-a] [-a] [-b SOURCE]
theHarvester: error: the following arguments are required: -d/--domain
(kali@kali)~$
$ theHarvester 4.10.1
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
* theHarvester
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usage: theHarvester [-h] -d DOMAIN [-l LIMIT] [-s START] [-p] [-s] [--screenshot SCREENSHOT] [-e DNS_SERVER] [-t] [-r [DNS_RESOLVE]] [-n] [-c]
                  [-f FILENAME] [-w WORDLIST] [-a] [-a] [-b SOURCE]
theHarvester: error: the following arguments are required: -d/--domain
```

```

kali@kali:~$ theHarvester -h
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
* theHarvester 4.10.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*****

usage: theHarvester [-h] -d DOMAIN [-l LIMIT] [-s START] [-p] [-s] [--screenshot SCREENSHOT] [-e DNS_SERVER] [-t] [-r [DNS_RESOLVE]] [-n] [--c]
                  [-f FILENAME] [-w WORDLIST] [-a] [-q] [-b SOURCE]

theHarvester is used to gather open source intelligence (OSINT) on a company or domain.

options:
  -h, --help            show this help message and exit
  -d, --domain DOMAIN    Company name or domain to search.
  -l, --limit LIMIT      Limit the number of search results, default=500.
  -s, --start START      Start with result number X, default=0.
  -p, --proxies          Use proxies for requests, enter proxies in proxies.yaml.
  -s, --shodan          Use Shodan to query discovered hosts.
  --screenshot SCREENSHOT Take screenshots of resolved domains specify output directory: --screenshot output_directory
  -e, --dns-server DNS_SERVER DNS server to use for lookup.
  -t, --take-over        Check for takeovers.
  -r, --dns-resolve [DNS_RESOLVE] Perform DNS resolution on subdomains with a resolver list or passed in resolvers, default False.
  -n, --dns-lookup       Enable DNS server lookup, default False.
  -c, --dns-brute        Perform a DNS brute force on the domain.
  -f, --filename FILENAME Save the results to an XML and JSON file.
  -w, --wordlist WORDLIST Specify a wordlist for API endpoint scanning.
  -a, --api-scan         Scan for API endpoints.
  -q, --quiet            Suppress missing API key warnings and reading the api-keys file.
  -b, --source SOURCE    Suppress missing API key warnings and reading the api-keys file.
                        dehashed, dnsdumpster, duckduckgo, fofa, fullhunt, github-code, gitlab, hackertarget, haveibeenpwned, hudsonrock, hunter,
                        hunterhow, intelx, leakix, leaklookup, netlas, onyphe, otx, pentesttools, projectdiscovery, rapiddns, robtex, rocketreach,
                        securitystorec3r, securitytrails, shodan, subdomainfinder, subdomaininderc99, thc, threatcrowd, tomba, urlscan, venacius,
                        virustotal, waybackarchive, whoissal, windvane, yahoo, zoomeye

```

```

kali@kali:~$ theHarvester -i
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
* theHarvester 4.10.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*****

usage: theHarvester [-h] -d DOMAIN [-l LIMIT] [-s START] [-p] [-s] [--screenshot SCREENSHOT] [-e DNS_SERVER] [-t] [-r [DNS_RESOLVE]] [-n] [-c]
                  [-f FILENAME] [-w WORDLIST] [-a] [-q] [-b SOURCE]
theHarvester: error: the following arguments are required: -d/--domain

kali@kali:~$ theHarvester -d tesla.com -b crtsh
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
* theHarvester 4.10.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*****

[*] Target: tesla.com
[*] Searching CRTsh.
[*] No IPs found.
[*] No emails found.
[*] No people found.
[*] No hosts found.

```

```

kali@kali:~$ theHarvester -d tesla.com -b crtsh -f tesla_report
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
* theHarvester 4.10.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*****

[*] Target: tesla.com
[*] Searching CRTsh.
[*] No IPs found.
[*] No emails found.
[*] No people found.
[*] No hosts found.

[*] Reporting started.
[*] XML File saved.
[*] JSON File saved.

kali@kali:~$ ls
demo      Downloads  file2.txt  index.html.1  myfolder2  Public  scan1.txt  tesla.json  tesla.xml
Desktop   example.txt  fruits.txt  Music         Pictures   sacn1.txt  scanme.nmap.org  tesla_report.json  Videos
Documents file1.txt    index.html  myfolder      Projects   sample.txt  Templates      tesla_report.xml

```

Experiment 2: Phishing Awareness Simulation Using GoPhish (20 Marks)

Aim

To conduct a phishing awareness simulation using GoPhish and analyze employee responses.

Tool Used

- Kali Linux
- GoPhish

Commands Executed

```
cd gophish
```

```
./gophish
```

Open browser:

```
https://127.0.0.1:3333
```

Campaign Report

Emails Sent : 100

Emails Delivered : 100

Emails Opened : 75

Phishing Link Clicked : 40

Credentials Submitted : 20

Attachment Downloaded : 10

Findings

The phishing simulation revealed that many employees interacted with the phishing email. Forty percent clicked the phishing link, while twenty percent submitted their login credentials. Several employees also downloaded a malicious attachment, demonstrating inadequate awareness of phishing attacks.

Security Weaknesses

- Employees trusted unknown email links.
- Login credentials were entered into fake websites.
- Suspicious attachments were downloaded.
- Lack of awareness about phishing techniques.
- Weak verification of email authenticity.

Recommendations

- Conduct regular phishing awareness training.

- Perform periodic phishing simulations.
- Enable Multi-Factor Authentication (MFA).
- Deploy secure email gateways.
- Configure SPF, DKIM, and DMARC.
- Encourage employees to report suspicious emails.
- Block executable attachments.
- Enforce strong password policies.

Conclusion

The GoPhish campaign demonstrated that phishing remains a major threat. Continuous employee awareness programs and stronger email security controls can significantly reduce successful phishing attacks.

Experiment 3: Nmap Aggressive Scan (-A) on scanme.nmap.org (10 Marks)

Aim:

To perform an aggressive scan on **scanme.nmap.org** using Nmap and analyze the discovered services.

Tool Used:

- Kali Linux
- Nmap

Commands used:

```
nmap --version
```

```
nmap -A scanme.nmap.org
```

```
nmap -A scanme.nmap.org -oN nmap_report.txt
```

```
ls
```

```
cat nmap_report.txt
```

OUTPUT:

```
(kali@kali)-[~]
└─$ nmap --version
Nmap version 7.99 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.8 openssl-3.6.2 libssh2-1.11.1 libz-1.3.1 libpcr2-10.46 libpcap-1.10.6 nmap-libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: epoll poll select

(kali@kali)-[~]
└─$ nmap -A scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 08:37 -0400
Stats: 0:00:26 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 25.00% done; ETC: 08:38 (0:00:15 remaining)
Stats: 0:00:32 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 50.00% done; ETC: 08:38 (0:00:11 remaining)
Stats: 0:00:39 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 50.00% done; ETC: 08:38 (0:00:18 remaining)
Stats: 0:00:50 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 50.00% done; ETC: 08:38 (0:00:29 remaining)
Stats: 0:01:00 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 50.00% done; ETC: 08:39 (0:00:39 remaining)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0023s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 985 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
22/tcp    open  tcpwrapped
|_ssh-hostkey: ERROR: Script execution failed (use -d to debug)
53/tcp    closed domain
80/tcp    open  http         Apache httpd 2.4.7 ((Ubuntu))
|_http-server-header: Apache/2.4.7 (Ubuntu)
139/tcp   closed netbios-ssn
143/tcp   closed imap
161/tcp   closed snmp
443/tcp   closed https
500/tcp   closed isakmp
993/tcp   closed imaps
3128/tcp  closed squid-http
3389/tcp  closed ms-wbt-server
5060/tcp  open  sip?
5061/tcp  open  sip-tls?
8080/tcp  closed http-proxy
8089/tcp  closed unknown
Device type: VoIP adapter|bridge|general purpose
Running (JUST GUESSING): AT&T embedded (90%), Oracle Virtualbox (88%), Slirp (88%), QEMU (86%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: AT&T BGW210 voice gateway (90%), Oracle Virtualbox Slirp NAT bridge (88%), QEMU user mode network gateway (86%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
```

```
(kali@kali)-[~]
└─$ nmap -A scanme.nmap.org -oN nmap_report.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 08:42 -0400
Stats: 0:00:21 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 41.35% done; ETC: 08:43 (0:00:24 remaining)
Stats: 0:00:34 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 25.00% done; ETC: 08:43 (0:00:15 remaining)
Stats: 0:01:19 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 75.00% done; ETC: 08:44 (0:00:17 remaining)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.032s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 988 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
22/tcp    open  tcpwrapped
|_ssh-hostkey: ERROR: Script execution failed (use -d to debug)
53/tcp    closed domain
80/tcp    open  http         Apache httpd 2.4.7 ((Ubuntu))
|_http-server-header: Apache/2.4.7 (Ubuntu)
139/tcp   closed netbios-ssn
143/tcp   closed imap
443/tcp   closed https
993/tcp   closed imaps
3128/tcp  closed squid-http
3389/tcp  closed ms-wbt-server
5060/tcp  open  sip?
5061/tcp  open  sip-tls?
8080/tcp  closed http-proxy
Device type: VoIP adapter|bridge|general purpose
Running (JUST GUESSING): AT&T embedded (90%), Oracle Virtualbox (88%), Slirp (88%), QEMU (86%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: AT&T BGW210 voice gateway (90%), Oracle Virtualbox Slirp NAT bridge (88%), QEMU user mode network gateway (86%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 0.40 ms scanme.nmap.org (45.33.32.156)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 110.47 seconds

(kali@kali)-[~]
└─$ ls
demo      Downloads  file2.txt  index.html.1  myfolder2  Projects  sample.txt  Templates  tesla_report.xml
Desktop  example.txt  fruits.txt  Music         nmap_report.txt  Public    scan1.txt  tesla.json  tesla.xml
Documents file1.txt    index.html  myfolder     Pictures    sacn1.txt  scanme.nmap.org  tesla_report.json  Videos
```

```
(kali@kali)~$ cat nmap_report.txt
# Nmap 7.99 scan initiated Tue Aug 4 08:42:25 2026 as: /usr/lib/nmap/nmap --privileged -A -oN nmap_report.txt scanme.nmap.org
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.032s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 988 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
22/tcp    open  tcpwrapped
|_ssh-hostkey: ERROR: Script execution failed (use -d to debug)
53/tcp    closed domain
80/tcp    open  http      Apache httpd 2.4.7 ((Ubuntu))
|_http-server-header: Apache/2.4.7 (Ubuntu)
139/tcp   closed netbios-ssn
143/tcp   closed imap
443/tcp   closed https
993/tcp   closed imaps
3128/tcp  closed squid-http
3389/tcp  closed ms-wbt-server
5060/tcp  open  sip?
5061/tcp  open  sip-tls?
8080/tcp  closed http-proxy
Device type: VoIP adapter|bridge|general purpose
Running (JUST GUESSING): AT&T embedded (90%), Oracle Virtualbox (88%), Slirp (88%), QEMU (86%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: AT&T BGW210 voice gateway (90%), Oracle Virtualbox Slirp NAT bridge (88%), QEMU user mode network gateway (86%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 0.40 ms scanme.nmap.org (45.33.32.156)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
# Nmap done at Tue Aug 4 08:44:16 2026 -- 1 IP address (1 host up) scanned in 110.47 seconds
```